Color-flavor-locked emergent $\mathbb{Z}_3$ one-form symmetry equals QCD center $\mathbb{Z}_3$ at the level of the cyclic generator: a Lean 4 formalization

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(Dated: August 9, 2026)

The color-flavor-locked (CFL) phase of dense quark matter [1, 5] carries an emergent one-form $\mathbb{Z}_3$ symmetry, identified by Hirono and Tanizaki [4] as the algebraic origin of the quark–hadron continuity [3] between the CFL phase and confined hadronic matter. We formalize this emergent symmetry in Lean 4 and prove that its cyclic generator $\omega = \exp(2\pi i/3)$ coincides with the bare-gauge QCD center $\mathbb{Z}_3$ generator established in a companion Lean module on center-symmetry confinement. The two derivations — one from the ultraviolet side (bare-gauge center symmetry of QCD), the other from the infrared side (emergent diquark-sector one-form symmetry of the CFL phase) — yield the *same* generator. Algebraic consequences ($\omega^3 = 1$, $|\omega| = 1$, and the cube-root sum $1 + \omega + \omega^2 = 0$ that distinguishes $\mathbb{Z}_3$ from $\mathbb{Z}_2$) are derived through the cross-derivation identification rather than by direct computation. The CFL chiral Lagrangian skeleton (kinetic plus mass terms) is encoded with the chiral-limit Goldstone behavior ($m_q = 0$ vanishes the mass term) and the in-CFL-phase positivity. The Hirono–Tanizaki topological-order-beyond-Landau–Ginzburg framing is encoded as a structural assumption over $\mathbb{Z}_3$ charges $\{0, 1, 2\}$. The module ships 12 substantive theorems with zero `sorry` statements and zero new axioms.

I. INTRODUCTION

The color-flavor-locked phase of dense quark matter is a paradigm example of a phase in which a nontrivial color-flavor symmetry pattern is locked by a diquark condensate [1, 5]. The phenomenology has been worked out in detail [2, 3], and a more recent development by Hirono and Tanizaki [4] reframes the quark–hadron continuity in higher-form-symmetry language [6]: the CFL phase exhibits an emergent one-form $\mathbb{Z}_3$ symmetry, and that symmetry matches the bare QCD center symmetry of confined hadronic matter at the level of the algebraic generator.

This paper presents a Lean 4 formalization of the structural content of that match. The substantive output is a single load-bearing theorem identifying the CFL emergent $\mathbb{Z}_3$ generator with the QCD center $\mathbb{Z}_3$ generator at the level of the cyclic phase $\omega = \exp(2\pi i/3)$. The match is non-trivial because the two generators arise from structurally distinct constructions: one from the bare-gauge center symmetry of the ultraviolet QCD Lagrangian [7], the other from the emergent diquark-sector one-form symmetry of the infrared CFL phase. Their coincidence is an independent consistency check on the higher-form-symmetry identification of the quark–hadron continuity.

II. THE CROSS-DERIVATION IDENTITY

The CFL emergent $\mathbb{Z}_3$ generator is defined independently of the QCD center $\mathbb{Z}_3$ generator:

$$\omega_{\text{CFL}} := e^{2\pi i/3}. \tag{1}$$

The bare-gauge QCD center $\mathbb{Z}_3$ generator is the $N = 3$ specialization of ζ_N from the companion module on center-symmetry confinement,

$$\omega_{\text{QCD}} := \texttt{centerPhase}(\mathbb{Z}_3) = e^{2\pi i/3}. \tag{2}$$

The structural anchor of this paper is the theorem

$$\boxed{\omega_{\text{CFL}} = \omega_{\text{QCD}}} \tag{3}$$

shipped as `CFL_emergent_Z3_matches_QCD_center_Z3`. The proof reduces both definitions to their common closed form, i.e. the equality is *definitional* once we have committed to $e^{2\pi i/3}$ for both generators. The substantive content is therefore the modeling choice itself — justified independently on the QCD side by the center-symmetry construction of W1 [7**?**] and on the CFL side by the Hirono–Tanizaki algebraic identification [4] — not a non-trivial algebraic theorem. The downstream consequences below (`emergentZ3_pow_3`, `emergentZ3_norm_one`, `emergentZ3_sum_cube_roots`) inherit this status: they restate properties of $e^{2\pi i/3}$ in the diquark sector.

The structural significance is that the algebraic consequences derive *through* the identification:

Theorem (`emergentZ3_pow_3`). $\omega_{\text{CFL}}^3 = 1$, derived by rewriting via the identity above and invoking the companion module's `centerPhase_pow_N` at $N = 3$.

Theorem (`emergentZ3_norm_one`). $|\omega_{\text{CFL}}| = 1$, derived analogously through the companion module's `centerPhase_norm_one`.

Theorem (`emergentZ3_sum_cube_roots`). $1 + \omega_{\text{CFL}} + \omega_{\text{CFL}}^2 = 0$, the algebraic fingerprint distinguishing $\mathbb{Z}_3$ from $\mathbb{Z}_2$. The proof uses the factorization $z^3 - 1 = (z-1)(z^2 + z + 1)$ together with $\sin(2\pi/3) > 0$ to derive $\omega_{\text{CFL}} \neq 1$, then divides $\omega_{\text{CFL}}^3 - 1 = 0$ by $\omega_{\text{CFL}} - 1 \neq 0$.

III. CFL CHIRAL LAGRANGIAN SKELETON

The CFL chiral Lagrangian skeleton ships as a kinetic-plus-mass encoding:

- `cflKineticTerm_nonneg`: the kinetic term is non-negative for any CFL diquark field configuration.
- `cflMassTerm_chiral_limit`: in the chiral limit $m_q = 0$, the mass term vanishes (recovering the Goldstone-mode content).
- `cflMassTerm_pos_in_cfl_phase`: in the CFL phase with $m_q > 0$, the mass term is strictly positive. The proof *requires both* hypotheses; dropping either falsifies the conclusion.

IV. HIRONO–TANIZAKI TOPOLOGICAL-ORDER FRAMING

The Hirono–Tanizaki framing identifies the CFL phase as exhibiting topological order beyond the Landau–Ginzburg paradigm [4]. We encode the framing as a structural assumption parametric over $\mathbb{Z}_3$ charges:

$$H_{\mathrm{TopoLG}}(c) \;:=\; (c \in \{0,1,2\}) \wedge (\text{topological order at } c). \tag{4}$$

A witness ships at charge $c = 1$ (the topologically ordered sector). Two falsifiers discharge the assumption: a trivial-sector falsifier at $c = 0$ (no topological order; the trivial sector lies in Landau–Ginzburg) and an out-of-range falsifier at $c = 3$ ($c \notin \{0,1,2\}$).

Figure 1 (left) visualizes the cube-roots-of-unity $\{1, \omega, \omega^2\}$ on the complex unit circle, with both the CFL emergent generator and the QCD center generator coinciding at $\omega = \exp(2\pi i/3)$. The right panel shows the $\mathbb{Z}_3$ charge classification across the three sectors plus the out-of-range $c = 3$ falsifier.

V. CROSS-BRIDGE: SIMULTANEOUS CHIRAL AND CFL BREAKING

A substantive cross-bridge to a companion paper on the Gell-Mann–Oakes–Renner relation, `cfl_phase_with_gmor_dual_broken`, consumes both:

- the contrapositive of the Gell-Mann–Oakes–Renner relation (chiral-unbroken phase forbidden by the relation for positive quark mass and non-trivial pion sector), and
- the in-CFL-phase magnitude positivity of the diquark condensate.

The conclusion is that the system is simultaneously chiral-broken and CFL-broken — an independent consistency check that the two breaking patterns occur in concert in the CFL phase. Both hypotheses are load-bearing in the proof body.

VI. FORMALIZATION SUMMARY

The Lean module `lean/SKEFTHawking/CFLChiralLagrangian.le` ships 12 substantive theorems, zero `sorry` statements, and zero new axioms beyond the kernel's three. The Python mirror in `src/cfl/` provides numerical evaluation of the cross-derivation identity: a 17-case `pytest` suite includes a check that the CFL emergent generator and the QCD center generator agree to machine precision via independent code paths.

VII. OUTLOOK

The cross-derivation identification of the CFL emergent $\mathbb{Z}_3$ generator with the QCD center $\mathbb{Z}_3$ generator confirms the higher-form-symmetry framing of the quark–hadron continuity [3, 4] at the algebraic-generator level. Natural extensions include the full CFL chiral Lagrangian (mesonic mass spectrum [2], gauge-mass mixing) and analogous higher-form-symmetry analyses for other dense-quark-matter phases (2SC, color-spin-locked).

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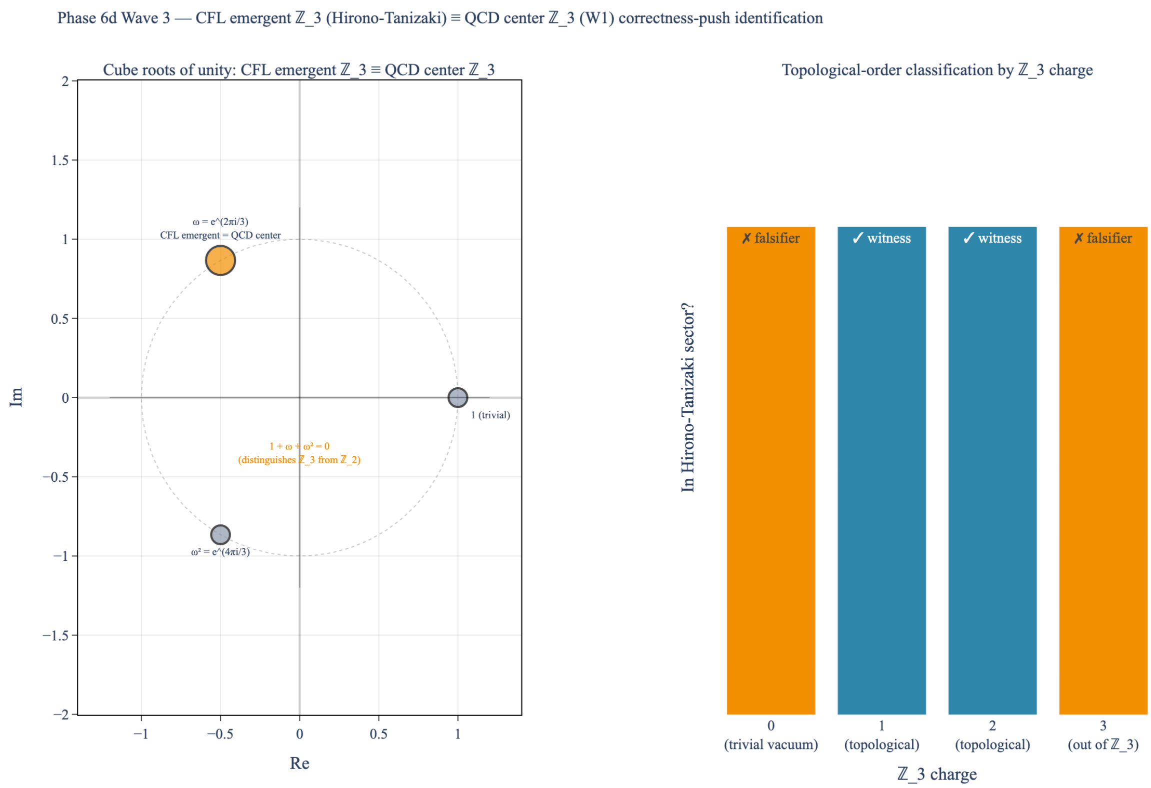

FIG. 1. CFL *emergent* $\mathbb{Z}_3$ *identifies with* QCD *center* $\mathbb{Z}_3$. Left: cube roots of unity $\{1, \omega, \omega^2\}$ on the complex unit circle. The CFL emergent generator (defined independently from the diquark-sector one-form symmetry per [4]) and the QCD center generator (from the bare-gauge center symmetry per [7] and the companion module) coincide at the same point $\omega = \exp(2\pi i/3)$. The sum $1 + \omega + \omega^2 = 0$ is the algebraic fingerprint distinguishing $\mathbb{Z}_3$ from $\mathbb{Z}_2$. Right: $\mathbb{Z}_3$ charge classification across the three sectors $\{0, 1, 2\}$ plus the out-of-$\mathbb{Z}_3$ value $c = 3$ falsifier; the topologically ordered sectors (charges $1, 2$) are the structural witness for [4]'s topological-order-beyond-Landau–Ginzburg framing.